Machine Learning Framework For Yield Prediction Using Soil Health And Climate Data

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***Abstract*—Under climate changing condition, sustainable agriculture and intelligent decision making is very necessary. Change in weather and soil distribution in the earth had made the work even tougher specially for the statistical procedures which are being used now for forecasting the yield of crops. In our model we had integrated climate and soil health for predicting the yield of the crops. For the project we have taken two crops i.e corn and soyabean and collected the climate and soil health dataset required for these crops. Our datasets contain of weekly weather data, characteristics of crops and soil at many levels. We have removed the non-informative features and zero-valued predominant features by using data preprocessing to make our model more robust. We have used many regression models like linear regression, decision tree, k-nearest neighbor, support vector regression, random forest, gradient boosting regression methods, ensemble regression and XGBoost. Out of all of these models XGBoost regression model was having the highest performance metrics, so we fine-tuned our model with that. We have built a platform where farmers can give input of the soil and climate health and they can get the predicted output of the yield of the crops.**

Keywords—crop yield prediction, XGBoost, Climate Data, Precision Agriculture, Machine Learning

I. INTRODUCTION

Agriculture had been an very essential factor of survival for humans and all the humans are directly dependent on the agricultural food items either directly or indirectly. And India being a agricultural dependent country where majority of the population in India are directly connected to agriculture as their primary and only source of income we have to focus on many of the innovative agricultural procedures. The innovation in crop yield is very important our farmers and for the population of the country to maximize the current resource utilization, overcome risks and provide a sustainable production of food. Yield of crops is however, a very tough and complicated process as there are many factors which are included in this like climate condition, soil quality and measures taken for management of the crops. There is a lot of uncertainty during the process of crop yield like the climate change, events of adverse weather etc. The conventional methods which are used for estimating the crop yield are either based on statistics or empirical relationship between the crop yields and earlier predictions. This approach provides basic insights but are not sufficient enough for doing high dimensionality predictions of agroclimatic variables, to be more specific weekly weather variation, type of soil at different depth and management decisions. This venerability needs a data-driven solution for big agricultural datasets. Additionally, the most previous studies done on the agricultural fields are either based on Climate or Soil health, but our study is to combine both the climate, soil and farming data for both the soyabean and corn datasets which contains weather data, soil data at different depth and planting week.

Comprehensive machine learning model [1] to predict the suitable crop and predict their yield using the agricultural dataset. The factors considered for the prediction are nutrients, rainfall, temperature, soil pH, season and cultivation area. Used many machine learning model to predict the best crop and also to predict the crop yield

Geospatial and machine learning model [2] used to identify the crop and calculate the yield of the crop. It uses satellite data which is collected overtime. It analysis the relationship between vegetation data and crop yield to identify the specific crop region and their stage of growth. It then uses the required regression model to estimate the yield based on the calculation

Stacking Ensemble Model (SEM) [3] and it's improved version OSEM to enhance the prediction. It uses real agricultural data along with artificial agricultural data generated by prophet algorithm. The model uses many machine learning algorithms and combines them to predict the result using a Extra-tree regressor. The function further optimizes the model using hyperparameter tuning and dimensionality reduction.

Multimodal and temporal deep learning framework [3] for crop yield classification by combining different types of data such as RGB images, infrared images, multispectral images and environmental stress data to classify the conditions for crop. It uses advance techniques such as combing the information from different types of data, reducing the complexity in data and how the stress effect the crops time to time.

Hybrid intelligent system [5] for analysing the soil and recommending crops with combines IoT and machine learning. A portable device that captures the real-time soil data like pH, nutrients and moisture present in the soil, it then suggests advice for crops in a understandable way.

II. Related Works

Machine learning based approach [6] for predicting the crop yield by using a technique known as Mutual Information Feature Selection [MIFS] with Multilayer Stacked Ensemble Regression [MSER] model. It includes methods such as weather condition, soil properties and practices for managing crops. The model uses many models which are stacked to improve the prediction ability.

Hybrid and smart climate framework [7] for predicting the crop yield by combing the traditional methods and machine learning approach. It combines many regression models like Random Forest, XGBoost by using a stacked ensemble technique to improve prediction. It also include a SHAP value which explains how different factors effect the yield of crops. It mainly focuses on climatic factors.

Data-driven approach [8] to predict the yield of crop using random forest and XGBoost. Uses historical agricultural and climate data which include rainfall, temperature, pesticide usage and area of land . Climate data using NASA powered API had been integrated.

Spatial-temporal deep learning model [9] using ConvLSTM architecture for estimation of yield of crop in difficult terrains. Uses satellite and SAR imaginary as source for data and include crop growth stage over a period of time.

Legume and other nutrients management [10] practice can decrease the pesticides remnant and increase the health of the soil in polluted agricultural fields. The outcome is to integrate combine legumes and biofertilizers notably lower pesticides level while increasing the soil nutrient which lead to overall productivity, making it a productive process for sustainable soil remedy.

Relationship between climate and yield of wheat using satellite data, climate information and machine learning. To better understand the conditions which affect the crop yield uses indicators like NDVI and EVI [11] along with environment and house data.

Sustainable agriculture and global food security [12] in the circumstance of climate change, while planning for better land and water management and increase in stress tolerant crops and reducing waste by using advance technology like remote sensing, IoT, AI and biotechnology to improve the agricultural system.

Analyse the trend of crop yield, climate change and economic condition in many agricultural zones [13] for farmers. By this it identifies the effect of climate on agriculture, for this farmers should adopt some smart practice for farming.

A total of 31,999 research papers [14] are analysed for the soil health. It shows many research is done in these plays in China, India, US and Spain. But regions of Africa and Southern Asia and America have less research. These ignored areas have a lot of problems like poor health of soil, climate change and lack of food.

Change in climate [15] had affected the agriculture which eventually impact the farmer's income in Italy, main focus on the grapes, fruits, vegetables and dairy farming by inspecting the changes measured in temperature, pattern of rainfall, and extreme weather condition.

Climate Smart Agricultural Practice (CSAPs) [16] impact the crop production and increase the risk for farmers growing rice in Afghanistan. Data shows different farming practice influence the productivity of crop at the same time time giving importance to training education and financial support to improve production

Deep Learning for prediction of crop yield called DLMWOA-CYPBCC [17]. Merge hybrid RNN-LSTM network and RFOA to improve the performance in prediction specially in climate change condition.

Small farmer perceive climate change [18] by merging the data of farmer survey and meteorological data to predict the trend in agriculture and climate by comprehending the effect of rainfall change and temperature pattern on the crops.

A large and diverse dataset [19] for crop monitoring in the United States based on the climate. Data contains the Satellite imagery, weather data, information of soil features of terrain and agricultural management data which can be used to predict the crop monitoring, prediction of yield and analysis of impact on climate.

III. Dataset Description

We are using large agricultural-climate dataset which contains two very important crops, namely corn and soyabean. Our dataset consists of three important features: Weather data, soil condition and factors for crop management. Climate data contains six components namely precipitation, solar radiation, snow water equivalent, maximum temperature, minimum temperature and vapor pressure. The soil conditions contains surface and deep soil properties which contain wet soil bulk density, dry bulk density, clay percentage, upper limit of plant available water content, lower limit of plant available water content, hydraulic conductivity, organic matter percentage, pH, sand percentage and saturated volumetric water content variables were observed at different depth. Crop management data include planting dates consisting of 16 components in terms of week.

IV. Data Pre-Processing

The raw data which we have obtained for the crops of corn and soyabean are high-order and diverse, which contains many variables representing weekly weather data, soil properties in multiple depths, and information of management of crops. Due to large data collection, the raw data includes a significant number of unnecessary, constant and inadequate features. To be more precise many weather and soil variables contains zero or near to zero values in many examination, which indicates to missing of meaningful variations for the particular parameters in the time period. These features can indicate noise, increase the computation difficulty and make a bad impact on the generalization of the model.

To overcome these challenges, a initial analysis was made to access the distribution of data, feature variance and fullness. The columns which contain zero variances were recognized and removed, because they don't contribute in the training. Many

features which have a zero value dominance beyond a predetermined threshold were eliminated to make sure that only the variables containing the information are maintained. By this process there is a reduction in the dimensionality while keep safe the critical important weather and soil related information. The remaining features were investigated and for consistency and numerical stability. By this preprocessing of the raw data, the dataset had been cleaned with good inputs which is very suitable for the training of the model and accurately predicting the crop yield.

V. System Design

The proposed system design is the complete architecture for the Climate aware crop yield prediction. It easily combines with machine leaning as a data-driven process with a User Interface with which the farmers can interact. The main goal is to estimate the crop yield for the farmers in which they can give their input details of the weather and soil.

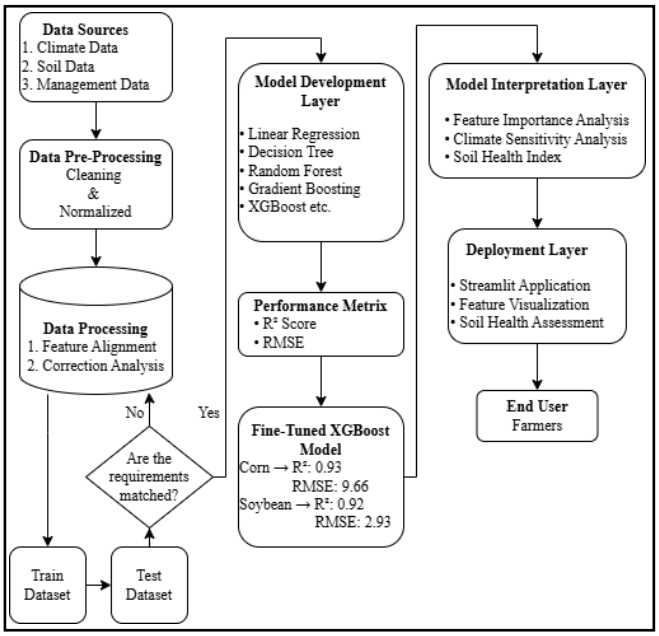


Fig. 1. *Architecture Diagram*

According to the architecture diagram, we have 2 datasets i.e corn and soyabean in each dataset it consists of 3 types of data i.e the Climate data, soil data and management data all of these are in the data source section. Once after data is collected it goes to the cleaning and the normalization process to remove noise. Then in the Data Processing section the feature alignment and correction analysis is done so that we can train the dataset. Once after the training, it is tested to ensure it matches the requirement, if no then we check for the Data Processing again, and if it matches our requirement we then go in the stage of model development where we try different types of model on the dataset like the linear regression, decision tree, random forest etc. then we try the Parformance matrix on the regression models for which we use the RMSE and the R^2 and based on the value we got the highest value in XGBoost. The next step is of fine-tuning where we fine-tuned both the datasets. Then we have done the Model Interpretation Layer where in which the Feature importance analysis, climate sensitivity analysis and soil health analysis is executed. Then is the development phase where we have used the Streamlit application for the prediction of feature visualization and soil health assessment for end user i.e Farmer.

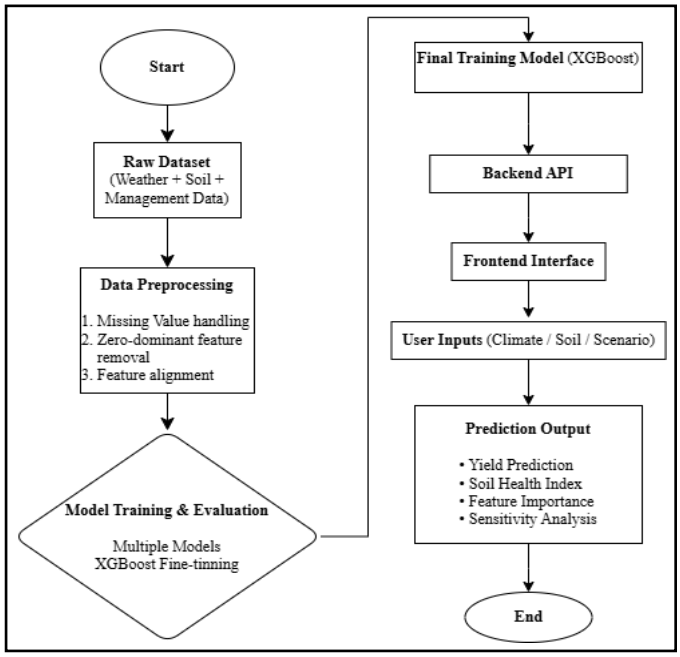


Fig. 2. *Dataflow Diagram*

VI. Proposed Work

A climate aware crop yield prediction system that uses agricultural and climate data along with machine learning technique and a website based interface. The main aim is to provide a system that can accurately predict the yield of the crop based on the a range of climate, soil and crop management system. The system works with two datasets of crops i.e the corn and the soyabean.

The step is to do data preparation, for this the main features in the raw dataset include whether data, soil data at different depth and crop management data. Cleaning of the data is to remove the zero dominant values for dimensionality reduction and noise removal. Then we go for data processing where we do feature extraction and train the model and test it, if the test result meets our satisfaction then we go for the model development stage and if it don't meets our requirement the data preprocessing is done again. In the model development stage we check our model performance matrix with different of the regression model like Linear Regression, Decision Tree, Random Forest, Support Vector Machine XGBoost and many more. Based on the calculations, XG boost was having the highest score in the performance matrix so the model was fine-tuned for both the crops dataset. The next step is of the Interpretation in which the feature importance analysis, climate sensitivity analysis and soil health index is performed on the dataset. At last, the system is deployed as an website based application in which the user can provide the input of the climate and soil type. By combining the machine-learning approach prediction is made for the yield of the crop.

VII. Result and Discussion

A. Experimental Setup

The dataset of corn and soyabean contains the components: weekly data, soil data and crop management data. After completion of the pre-processing step the dataset is spitted for training and testing of the model's performance.

Model performance are calculated by using the two performance matrix for regression:

- Root Mean Square Error (RMSE)

$$RMSE = \sqrt{\sum_{j=1}^{n} \frac{(x_j - \hat{x})^2}{N}} \quad (1)$$

where √ is the square root. N is the total number of observations or data points in the data set. x_j is the original value of x. $\hat{x}$ is the predicted value of x. $\sum_{j=1}^{n} \frac{(x_j - \hat{x})^2}{N}$ is the squared summation of absolute difference in the actual and predicted values ranges from j = 1 to n, which is divided by N.

- R^2 (R-Squared)

R-squared is also termed as Coefficient of determination. It represents the closest determination of coefficients to the original values. R-Squared values lie between 0 and 1 where 0 represents the model doesn't fit with the dataset, 1 represents the model perfectly fit with the given data.

$$R^2 = 1 - \frac{\sum(x_j - \hat{x})^2}{\sum(x_j - \bar{x})^2} \quad (2)$$

where x_j is the original value of x. $\hat{x}$ is the predicted value of x, $\bar{x}$ is the mean value of x. N is the total number of observations or data points in the dataset, $\sum(x_j - \hat{x})^2$ is the square of the difference in the actual and predicted values. $\sum(x_j - \bar{x})^2$ is the square of the difference in the actual values and mean value of x.

B. Discussion

The result shows the ensembling model had higher performance matrix value compared to linear regression. Decision Tree has the lowest performance matrix value because of overfitting and less generalization volume for large dimensional dataset.

Multiple Linear regression has a higher performance matrix value which shows a notable existence of linear data. Ensemble model, Random Forest, Gradient Boosting and XGBoost had done better when it comes to resilience and consistency.

The XGBoost model had the highest performance matrix after fine-tuning for both the crops dataset which contains the lowest RMSE value and highest R^2 value. This result shows the ability of the model nonlinear relationships and collaboration between the weather data, soil data and crops management data. The system gives the feature importance analysis and climate sensitive visualization, by this the transparency and usability is increased.

C. Yield Distribution Analysis

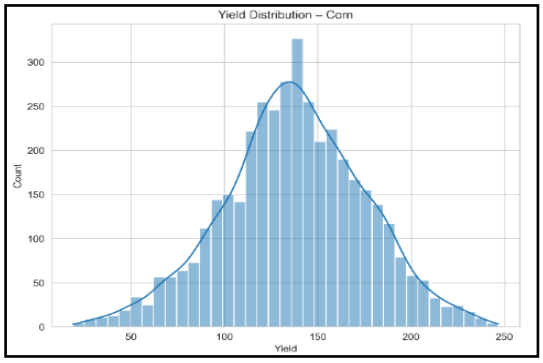


Fig. 3. *Yield distribution of Corn Dataset*

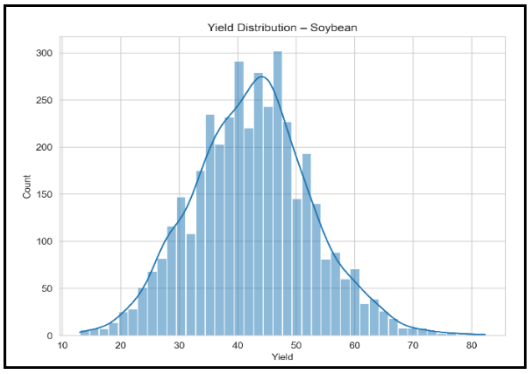


Fig. 4. *Yield distribution of Soyabean Dataset*

Fig. 4. and Fig. 5. shows the yield distribution vs count for both the corn and the soyabean datasets. The histogram shows both the datasets are roughly bell shaped distribution which means the values for the yield are continuous and spread in a observable range. In the corn dataset the yield distribution is centred and have a moderate production which is lopsided where the yield is higher which indicates change in climate and soil conditions in different seasons. Soyabean yield has a slim distribution which means a stable production when using similar agricultural and climate conditions. The missing of extreme outliners proves the preprocessing feature cleaning steps are effectively removed values.

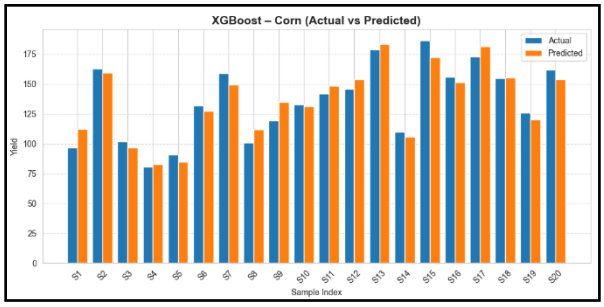


Fig. 5. *Actual VS Predicted Yield distribution of Corn Dataset using XGBoost*

The performance of the XGBoost in prediction of yield is consistent with actual yield distribution as shown in Fig. 5. and Fig. 6.

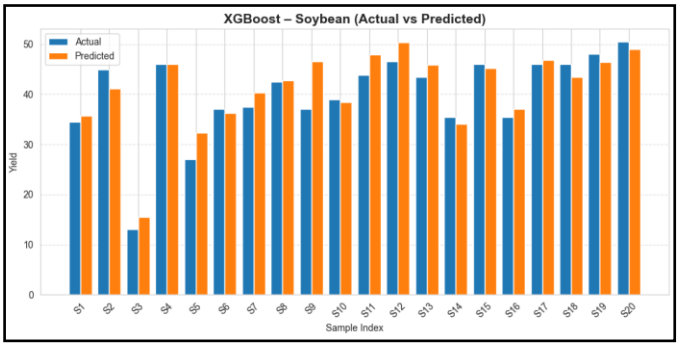


Fig. 6. *Actual VS Predicted Yield distribution of Soyabean Dataset using XGBoost*

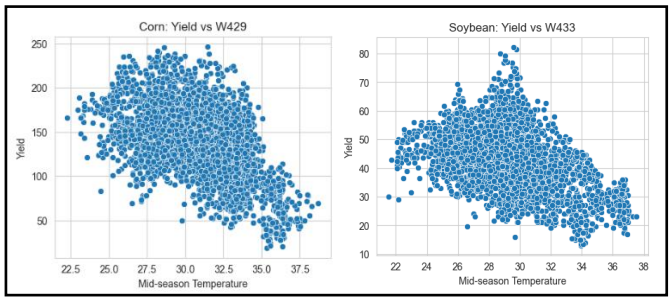


Fig. 7. *Relationship between mid season temperature (W429) and Corn Yield. And relationship between mid-season temperature (W433) and Soyabean yield*

Fig. 8 shows the scatter plot means a non-linear relationship between the temperature and corn yield, with mid-season temperature with high yield and very high temperature with low yield levels

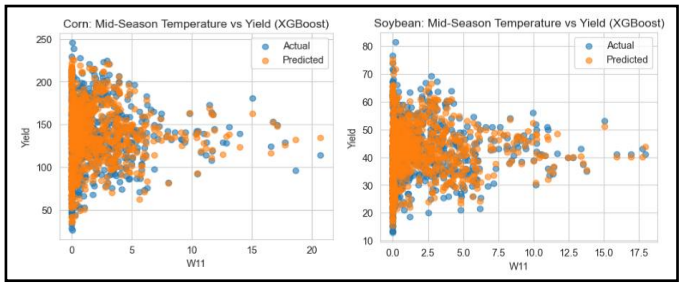


Fig. 8. *Actual VS Predicted Temperature Relation (XGBoost regression)*

In Fig. 9. the XGBoost and the Gradient Boosting predict fast values which are near in relation to original temperature and indication of yield. This highlights the temperature is a viral feature in the climate condition for the yield of crops and support the ability of the model to do more accurately interact with the climate yield.

In Fig. 10. the corn variables W429, W428 and W431 shows a high negative correlations, which means high temperature at critical growth periods are harmful to productivity. Similarly, Soyabean yield is highly reactive to temperature-with related weather variable, with mainly negative connections in the top-ranked variables. The positive connection with some soil components and ability to recall abilities make a positive influence on the performance of the yield. This justify the choice of XGBoost as the final predictive model.

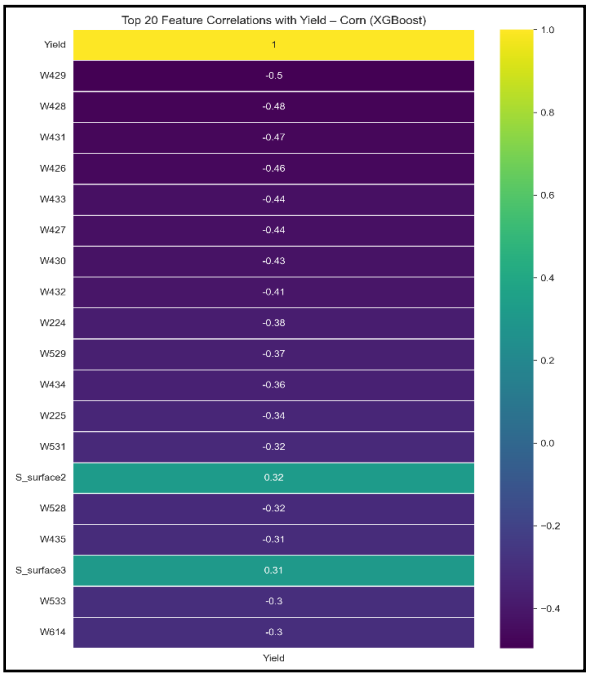


Fig. 9. *Top 20 feature correlation with yield – Corn (XGBoost)*

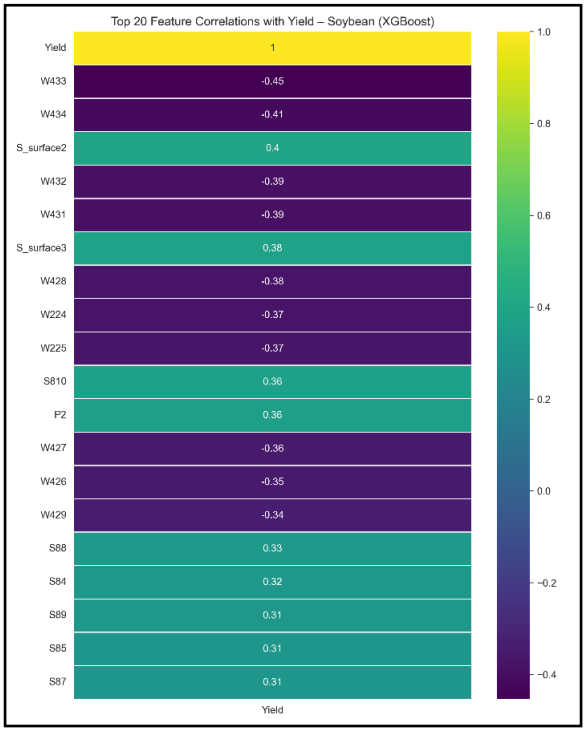


Fig. 10. *Top 20 feature correlation with yield – Soyabean (XGBoost)*

The heatmaps in Fig. 12. and Fig. 13. shows the ordered correlation pattern of many weather data for a period of weeks and soil related variables, which discover the presence of interdependency in the soil depth variables. Although the interdependency between the variables and yield are average,

the general relationship between the weather and soil variables improves the accuracy.

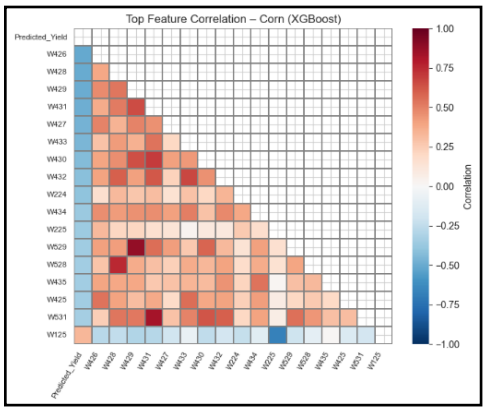


Fig. 11. Top Features Correlation heatmap – Corn dataset (XGBoost)

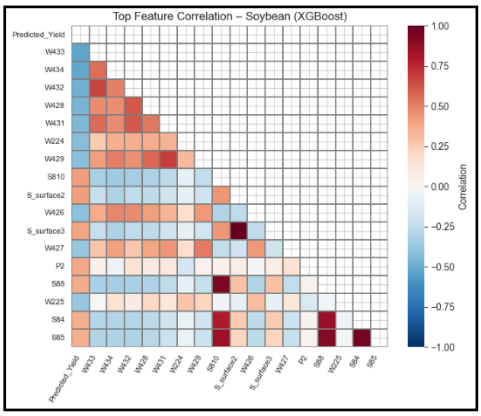


Fig. 12. Top Features Correlation heatmap – Soyabean dataset (XGBoost).



Fig. 13. User Interface for Soyabean in a Normal Year Scenario



Fig. 14. User Interface for Corn in a Hot Year Scenario

D. Model Comparison

Table I shows the performance matrix for the different regression models used on the dataset of corn and soyabean.

TABLE I. TABLE TYPE STYLES

Crop	***Model***	***RMSE***	***R^2***
Corn	Multiple Linear Regression	10.37	0.924
Corn	Decision Tree Regression	18.21	0.765
Corn	KNN Regression	13.22	0.876
Corn	SVR (Linear)	11.45	0.912
Corn	Gradient Boosting Regression	11.81	0.901
Corn	Ensembling	10.01	0.929
Corn	XGBoost Regression	9.88	0.931
Corn	XGBoost (Fine-Tuned)	9.66	0.934
Soyabean	Multiple Linear Regression	3.11	0.908
Soyabean	Decision Tree Regression	5.26	0.736
Soyabean	KNN Regression	3.94	0.852
Soyabean	SBR (Linear)	3.49	0.932
Soyabean	Gradient Boosting Regression	3.55	0.880
Soyabean	Ensemble Regression	3.07	0.910
Soyabean	XGBoost Regression	2.94	0.918
Souabean	XGBoost (Fine-Tuned)	2.93	0.918

VIII. FUTURE IMPROVEMENTS AND CONSLUSION

The present work point of focus is mainly on the phenotypic factors such as the climate, conditions of soil and management practices. In the future studies, the model can be intensified by adding the genotypic data of the crops for a better understanding of variations in the genetics that can impact the yield of the crops.

Climate-aware crop yield prediction system which is using machine learning technique on the two datasets i.e Corn and Soyabean in which the input data can be divided into a number of components with respect to weather, soil, and management data.. Many models of regression are evaluated and the XGBoost shows the highest R^2 and lowest RMSE value. There is an improvement in the explainability of the analysis of feature importance. The interface of the model is accessible and user-friendly. The approach gives a practical solution for the making of decision at the time of various conditions of the change in climate.

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